

2016.0 RANGE ROVER (LG), 206-05

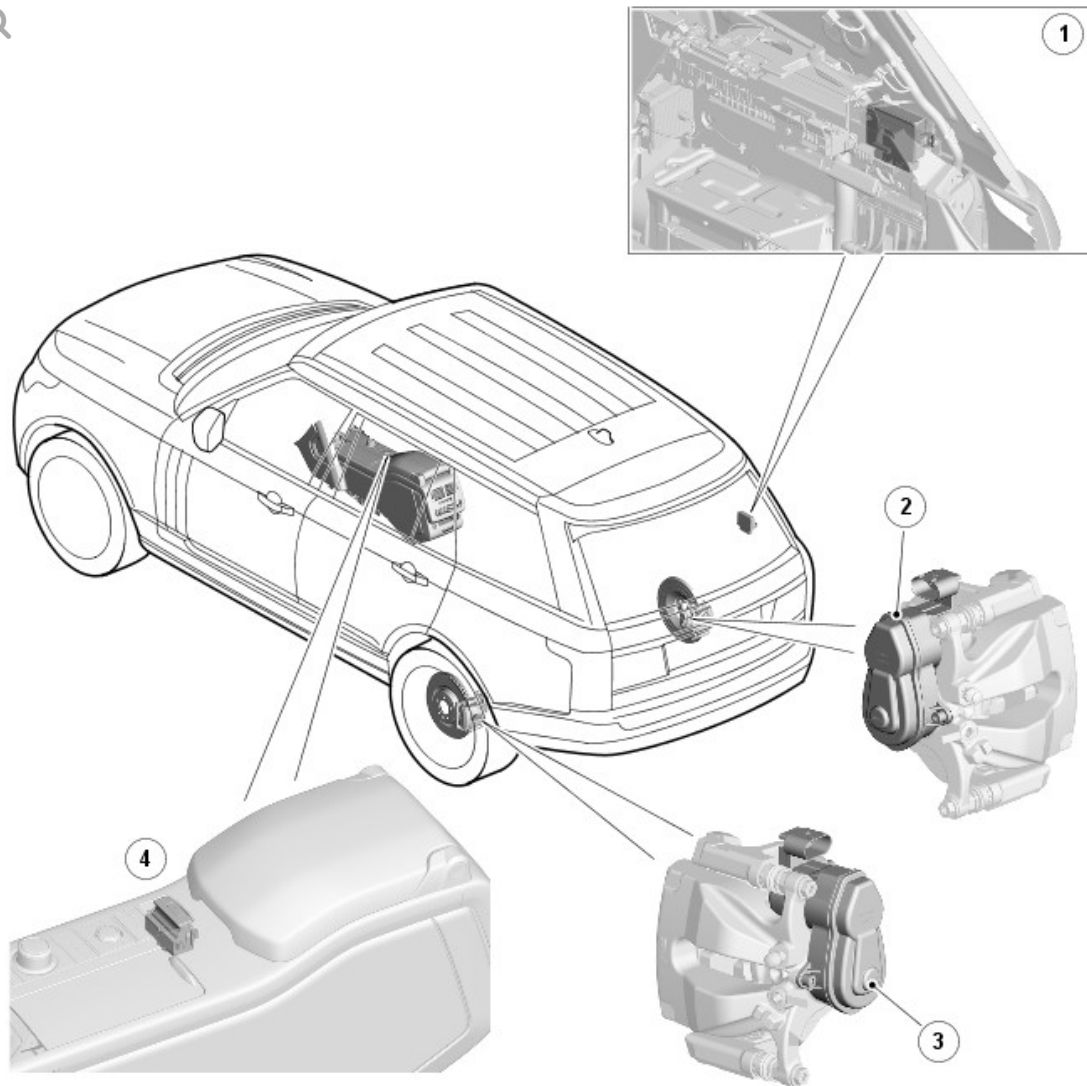
PARKING BRAKE AND ACTUATION

DESCRIPTION AND OPERATION

COMPONENT LOCATION

NOTE:

RHD (right-hand drive) installation shown, LHD (left-hand drive) installation similar.



E141858

ITEM	DESCRIPTION
1	Electric park brake control module
2	Right electric park brake actuator
3	Left electric park brake actuator
4	Electric park brake switch

OVERVIEW

The EPB (electric park brake) is an electrically actuated system that operates the calipers on the rear brakes. The system consists of:

- An EPB switch

- Two EPB actuators
- An EPB control module
- Two warning indicators.

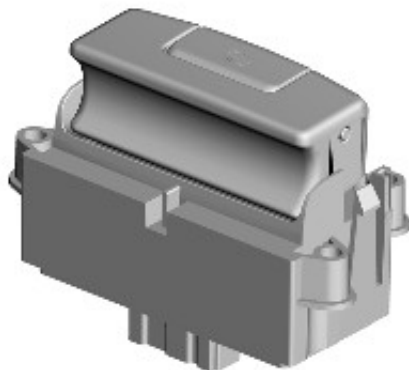
The EPB is controlled by the EPB control module, which uses the EPB actuators to apply and release the rear disc brakes. Operation of the EPB is initiated by the EPB switch and software within the EPB control module. The EPB has the following operating modes:

- Static apply/release
- DAR (drive away release)
- Automatic transmission release
- High temperature re-clamp
- Driver exit warning
- Dynamic apply.

Signals from other vehicle systems for operation of the EPB are transmitted to the EPB control module on the high speed CAN (controller area network) powertrain bus.

DESCRIPTION

ELECTRIC PARK BRAKE SWITCH



E141859

The park brake is manually applied and released using the EPB switch, which is located in the floor console. Pulling the switch lever upwards applies the park brake. Pushing the switch lever down releases the park brake.

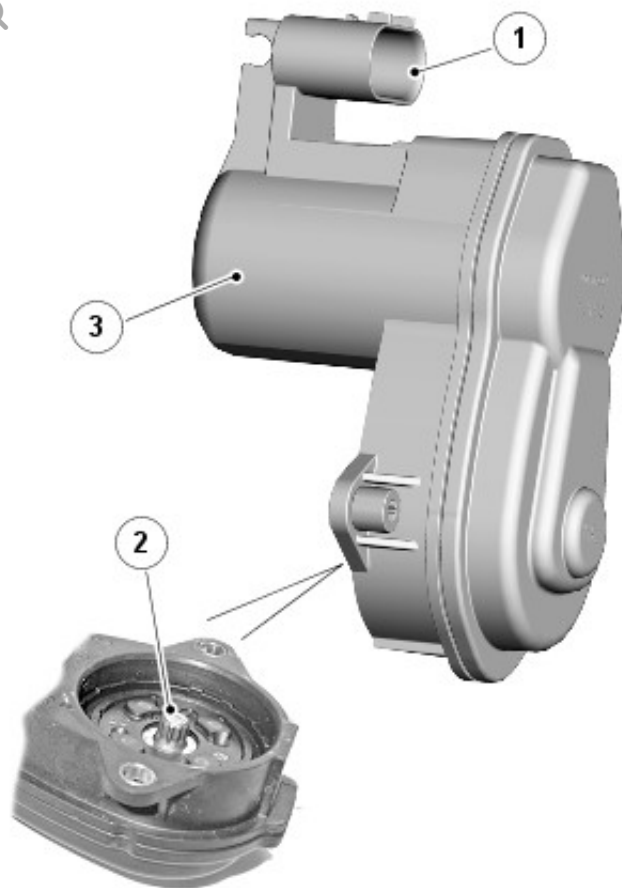
The resistive ladder type switch has four hardwired connections with the EPB control module, consisting of two apply circuits and two release circuits. This enables the EPB control module to perform a plausibility check on each switch selection.

Interior lighting switch illumination is provided by a hardwired connection with the CJB (central junction box).

The switch also incorporates two hardwired connections with the terrain response switch pack for use by the terrain response system.

For additional information, refer to: [Ride and Handling Optimization](#) (204-06 Ride and Handling Optimization, Description and Operation).

ELECTRIC PARK BRAKE ACTUATORS

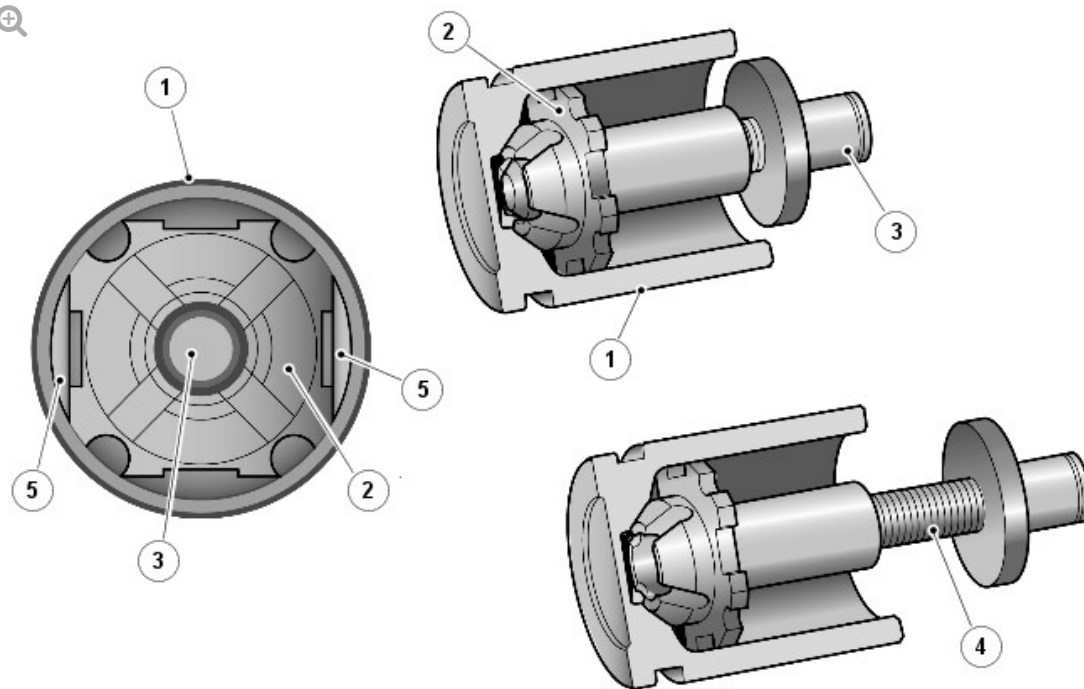


E141860

ITEM	DESCRIPTION
1	Electrical connector
2	Output shaft
3	Motor

The EPB actuators are electro mechanical positioning units attached to the calipers of the rear brakes. Each actuator incorporates a reversible electric motor driving an output shaft via a drive belt and planetary gear cluster.

The output shaft of the EPB actuator is engaged in the splined end of a spindle, which is inside the caliper piston and screwed into a thrust nut. Operation of the electric motor turns the spindle and produces linear movement of the thrust nut, to apply a target clamp load to the brake pads.



E133538

ITEM	DESCRIPTION
1	Caliper piston
2	Thrust nut
3	Spindle
4	Spindle thread
5	Flat surface

Two flat surfaces on the inside of the caliper piston prevent the thrust nut from turning with the spindle. Scallops around the circumference of the thrust nut allow the free flow of hydraulic fluid around the nut during mechanical operation by the EPB system and hydraulic operation by the brake pedal or ABS (anti-lock brake system) control module.

Operation of each EPB actuator is controlled by the EPB control module via two hardwired connections with the electric motor. The EPB control module changes the polarity of the connections to change the direction of drive of the motor.

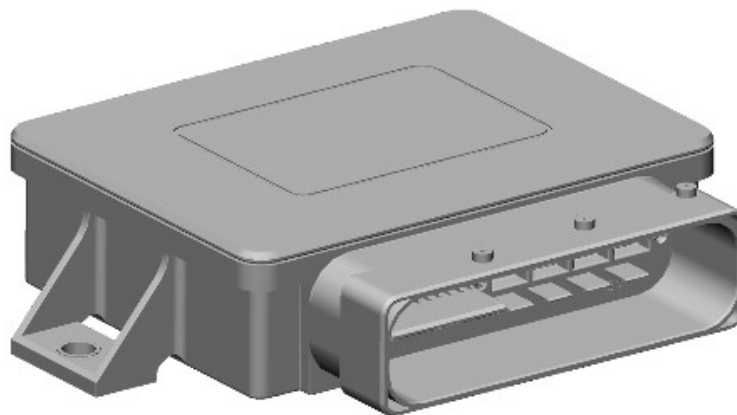
In each EPB actuator, when the park brake is applied the electric motor

turns the spindle to press the thrust nut against the caliper piston and apply the brake. The EPB control module measures the current draw of the electric motor and disconnects the power supply when it equates to a clamp load of 18.5 kN being applied by the thrust nut.

Each EPB actuator incorporates a temperature sensor which is hardwired to the EPB control module. The actuator temperature affects the clamp load applied for a given current draw, so the EPB control module uses the temperature sensor inputs to constantly adjust the cutoff current value, to ensure a minimum clamp load is always achieved.

When the park brake is released, the electric motor turns the spindle for approximately 800 milliseconds to retract the thrust nut and release the pressure on the caliper piston.

ELECTRIC PARK BRAKE CONTROL MODULE



E133209

The EPB control module is located behind the right rear quarter trim panel in the loadspace.

The EPB control module controls the two EPB actuators using a dual micro controller to provide separate control transmission for each actuator. An inclination/ gravity sensor is integrated into the EPB control module. The sensor allows the EPB control module to determine the longitudinal attitude (tilt angle) of the vehicle and vehicle deceleration.

Separate battery power feeds for the EPB actuators are supplied to the EPB

control module from the RJB (rear junction box). A further power feed is supplied to the EPB control module from the ignition relay in the CJB.

Four hardwired connections with the EPB switch provide the EPB control module with park brake apply and release inputs. Inputs from other components are received on the high speed CAN powertrain bus, as follows:

- Wheel speed sensors, from the ABS control module
- APP (accelerator pedal position) sensor, from the ECM (engine control module)
- Brake pedal switch, from the CJB
- Gear position, from the TCM (transmission control module).

In addition to the hardwired outputs to the EPB actuators, the control module outputs status and fault signals on the high speed CAN powertrain bus to operate the warning indicators and advise other systems if there is a fault with the EPB system.

WARNING INDICATORS



ITEM	DESCRIPTION
1	Park brake warning indicators (all except Canada, Mexico and USA)
2	Park brake warning indicators (Canada, Mexico and USA)

The EPB control module requests the display of the warning indicators and messages in the instrument cluster using high speed CAN powertrain bus signals.

When the park brake is applied in the static mode, the red park brake warning indicator is permanently illuminated while the ignition is on, and for 10 seconds after the ignition goes off. If a fault is detected and the park brake is unable to apply the full clamp load, the red park brake warning indicator flashes. If any other EPB fault is detected, the amber park brake warning indicator is illuminated. All fault indications are accompanied by a warning chime.

The red or amber park brake warning indicators, or the triangular red (critical) or amber (general warning/information) warning indicators, are also displayed in the message center, together with the EPB related messages. For additional information, refer to: [Instrument Cluster](#) (413-01 Instrument Cluster, Description and Operation).

OPERATION

When the park brake is applied with the EPB switch, the EPB control module uses the wheel speed inputs from the ABS control module to determine when to use the static and dynamic modes of operation. It uses the static mode at vehicle speeds up to 2 mph (3 km/h) and the dynamic mode at vehicle speeds greater than 2 mph (3 km/h).

STATIC APPLY/RELEASE

For the static apply to operate, the ignition can be on or off. When the EPB switch is pulled up, the EPB control module activates both EPB actuators to apply the brakes and signals the instrument cluster to turn on the red park brake warning indicator. The warning indicator remains on continuously

while the ignition is on, and for 10 seconds after the ignition goes off.

For the static release to operate, the ignition must be on and the brake pedal pressed. When the EPB switch is pressed down, the EPB control module then activates both EPB actuators to release the brakes and signals the instrument cluster to turn off the red park brake warning indicator.

DRIVE AWAY RELEASE

With the vehicle static and the park brake applied, the DAR (drive away release) mode gradually reduces the clamp load when the accelerator pedal is pressed, without the EPB switch being set to the release position, to assist with a smooth drive away. This mode is only active if either the driver door is closed, or if the driver safety belt is buckled and the engine is running.

The point at which the EPB control module releases the brakes is determined by the following factors:

- Tilt angle - from the inclination/ gravity sensor in the EPB control module
- Engine torque - from the ECM
- Throttle pedal applied - from the APP sensor
- D (drive) or R (reverse) selected - from the TCM.

DAR operation can be delayed by holding the EPB switch in the applied position, then releasing it at the required time.

AUTOMATIC TRANSMISSION RELEASE

The park brake is automatically released when the transmission control switch is taken out of P, provided either the driver door is closed, or the driver safety belt is buckled, the brake pedal is applied and the engine is running.

HIGH TEMPERATURE RE-CLAMP

After heavy use of the brakes the disc temperatures can be extremely high. At high brake disc temperatures, if the vehicle is stationary and the park brake is applied, the EPB control module automatically re-applies the park

brake as the brake discs cool down. A temperature model in the EPB control module software calculates the brake disc temperature from brake application force, the time the brake force is applied and vehicle speed.

High temperature re-clamp occurs at brake disc temperatures of 300 °C (572 °F) and above. Depending on the temperature up to three re-clamps can occur, at varying time intervals, in the 10 minutes following the initial application of the park brake. To limit the number of re-clamps, the ignition status as well as the current inclination of the vehicle is considered. If the ignition is on, a re-clamp will only be performed if the vehicle inclination is greater than 10%.

DRIVER EXIT WARNING

If the driver attempts to exit the vehicle with the park brake off and the ignition on or the engine running, the following warning signals occur when the driver door opens:

- The red park brake warning indicator illuminates
- A warning chime sounds
- A message center warning is displayed.

ROLL AWAY DETECTION

The park brake is automatically applied if the vehicle is stationary with the park brake off and then begins to move without operation of the accelerator pedal, indicating there is no driver intention to pull away.

DYNAMIC APPLY

While the vehicle is moving, the dynamic apply mode provides emergency braking. At a vehicle speed greater than 2 mph (3 km/h), pulling up and holding the EPB switch produces a gradual deceleration of the vehicle. The red park brake warning indicator illuminates, accompanied by a warning in the message center and a warning chime. The vehicle stop lamps also illuminate, activated by a CAN bus message from the EPB control module to the CJB. Dynamic apply operation is cancelled if the EPB switch is either released or pressed down to the release position, or the accelerator pedal is

pressed.

There are four modes of dynamic apply, which the EPB control module uses in the following order of preference:

- 1** ECD (electronic controlled deceleration)
- 2** RWU (rear wheel unlocker)
- 3** DSD (dynamic standstill detection)
- 4** DSD+.

The ECD mode uses hydraulic braking on all four wheels to decelerate the vehicle. The other three modes all use the EPB actuators to operate the rear brakes to decelerate the vehicle.

On HEVs (hybrid electric vehicles), the ABS control module can also use regenerative braking to help decelerate the vehicle. If regenerative braking is active when ECD mode is activated, the ABS control module maintains the same level of regenerative braking and uses hydraulic application of the brakes to produce any additional braking effort that is required.

ELECTRONIC CONTROLLED DECELERATION

The braking operation is controlled by the ABS control module, in response to a request from the EPB control module on the high speed CAN powertrain bus. The ABS control module monitors the deceleration rate using the wheel speed sensor signals, and adjusts the hydraulic pressure to the brakes as necessary to achieve the required deceleration. All of the brake control system operating modes remain enabled. As the vehicle comes to a halt, the hydraulic pressure (and/or regenerative braking on HEVs) is reduced to zero and the EPB control module applies the park brake with the EPB actuators.

REAR WHEEL UNLOCKER

The RWU mode is used if the ABS control module cannot operate the brakes or achieve a minimum deceleration rate, but still supplies wheel

speed signals to the EPB control module. The wheel speed inputs allow the EPB control module to adjust the clamp load on the rear brakes as necessary to produce maximum deceleration without locking the wheels. When vehicle speed decreases to 2 mph (3 km/h), the EPB control module fully applies the park brake.

DYNAMIC STANDSTILL DETECTION

The DSD mode is used if the ABS control module cannot operate the brakes or achieve a minimum deceleration rate, and the EPB control module has no valid wheel speed signals. In the DSD mode, the EPB control module uses the inclination/ gravity sensor to monitor deceleration. To maintain vehicle stability, the EPB control module initially applies a low clamp load to the brakes, then increases the clamp load in steps. When the inclination/ gravity sensor signal indicates the vehicle has stopped, the EPB control module fully applies the park brake.

DYNAMIC STANDSTILL DETECTION +

The DSD+ mode is used if the ABS control module cannot operate the brakes or achieve a minimum deceleration rate, the EPB control module has no valid wheel speed signals and the inclination/ gravity sensor has failed. In the DSD+ mode, the EPB control module applies a clamp load calculated to give a steady rate of deceleration without locking the wheels.

SERVICE

FUNCTIONAL TEST

The EPB control module incorporates a vehicle inspection mode that allows a functional test of the park brake to be performed on a rolling brake tester.

BRAKE PAD REPLACEMENT

To replace the rear brake pads or discs it is necessary to fully retract the thrust nuts to allow retraction of the caliper pistons.

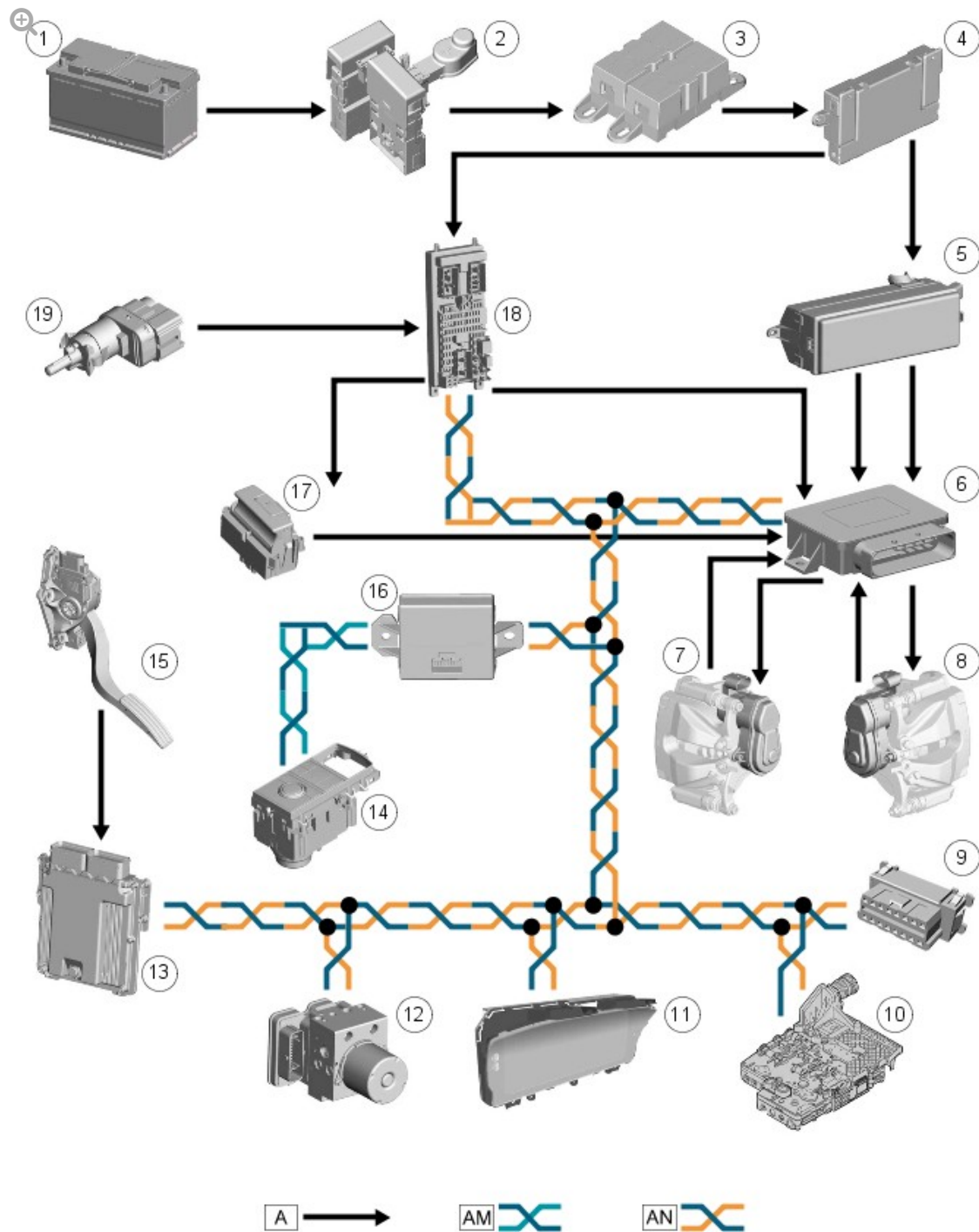
For additional information, refer to: [Electronic Parking Brake Service Mode Activation and Deactivation](#) (206-05 Parking Brake and Actuation, General

Procedures) /

Electronic Parking Brake Release When the Vehicle Has No Electrical Power

(206-05 Parking Brake and Actuation, General Procedures).

CONTROL DIAGRAM



E151889

A = HARDWIRED; AM = HIGH SPEED CAN CHASSIS BUS, AN = HIGH SPEED CAN POWERTRAIN BUS.

ITEM	DESCRIPTION
1	Battery
2	Megafuse
3	Battery Junction Box 2 (BJB2)
4	Battery Junction Box (BJB)
5	Rear Junction Box (RJB)
6	Electric Park Brake Control Module (EPBCM)
7	Left electric park brake actuator
8	Right electric park brake actuator
9	Diagnostic connector
10	Transmission Control Module (TCM)
11	Instrument Cluster (IC)
12	Anti-lock brake system control module (ABS)
13	Engine Control Module (ECM)
14	Terrain Response™ switchpack
15	Accelerator Pedal Position Sensor (APP)
16	Gateway Module (GWM)
17	Electric park brake switch
18	Central Junction Box (CJB)
19	Brake pedal switch